ENTERPRISE BRANCH NETWORK

Design, Configuration, Verification & Troubleshooting

CCNA Practical Project | Cisco Packet Tracer

Prepared by: Gautam Londhe

Target roles: L1 Network Engineer / NOC Engineer / Network Support Engineer

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1. Project Summary

This project designs a small enterprise branch network for ABC Technologies. It demonstrates VLAN segmentation, 802.1Q trunking, router-on-a-stick inter-VLAN routing, DHCP, single-area OSPF, PAT, an extended ACL, switch port security, SSH management, and structured troubleshooting.

Important scope note: the WAN-side test network is a simulated external network, not the public internet. A successful test to the simulated external web server demonstrates routing and PAT in the lab only.

1.1 Requirements

- Sales users belong to VLAN 10; Engineering users belong to VLAN 20.
- The server farm belongs to VLAN 99 and uses static addressing.
- R1 provides DHCP to Sales and Engineering.
- R1 and R2 form an OSPF Area 0 adjacency over a /30 point-to-point link.
- Internal hosts use PAT through R1's WAN interface to reach the simulated external server network.
- Sales must not initiate IP traffic to the server farm; Engineering is permitted.
- Access ports use port security; device management uses SSHv2.

2. Topology and Physical Port Plan

Use the following exact port map to avoid the invalid/nonexistent Gi0/0 and Gi0/3 assumptions on a typical Catalyst 3560. Interface names can vary by Packet Tracer model; if your selected device differs, map the equivalent available ports consistently.

Device / Port	Connects to	Mode / purpose
R1 Gi0/0	SW-CORE Fa0/24	802.1Q trunk; VLANs 10, 20, 99
R1 Gi0/1	R2 Gi0/1	WAN point-to-point link
R2 Gi0/0	External test server network	203.0.113.0/24 simulated external LAN
SW-CORE Gi0/1	SW1 Gi0/1	Trunk; VLANs 10, 20, 99
SW-CORE Gi0/2	SW2 Gi0/1	Trunk; VLANs 10, 20, 99
SW-CORE Fa0/23	Internal server	Access VLAN 99
SW1 Fa0/1-2	Sales PCs	Access VLAN 10
SW2 Fa0/1-2	Engineering PCs	Access VLAN 20
External server	R2 external LAN	Static IP 203.0.113.10/24, gateway 203.0.113.1

Logical path: Sales/Engineering PCs → access switch → core switch → R1 router-on-a-stick → R2 simulated WAN router → external test server.

3. IP Addressing Plan

Segment	VLAN	Network / mask	Gateway / address
Sales	10	192.168.10.0/24	192.168.10.1 (R1 Gi0/0.10)
Engineering	20	192.168.20.0/24	192.168.20.1 (R1 Gi0/0.20)
Server farm	99	192.168.99.0/24	192.168.99.1 (R1 Gi0/0.99); server 192.168.99.10
R1–R2 WAN	—	200.1.1.0/30	R1 200.1.1.1; R2 200.1.1.2
Simulated external LAN	—	203.0.113.0/24	R2 203.0.113.1; test server 203.0.113.10

The 203.0.113.0/24 range is reserved for documentation and is used here only as a simulated external network.

4. Configuration

4.1 SW-CORE — Catalyst 3560

```
enable
configure terminal
hostname SW-CORE
no ip domain-lookup
vtp domain ABC-BRANCH
vtp password cisco123
vtp mode server

vlan 10
 name SALES
vlan 20
 name ENGINEERING
vlan 99
 name SERVER_FARM

! Trunk to R1
interface fastEthernet0/24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,99
 no shutdown

! Trunks to access switches
interface range gigabitEthernet0/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,99
 no shutdown

! Internal server access port
interface fastEthernet0/23
 switchport mode access
 switchport access vlan 99
 spanning-tree portfast
 no shutdown

spanning-tree vlan 10,20,99 root primary
end
copy running-config startup-config
```

Note: On some Packet Tracer 3560 images, `switchport trunk encapsulation dot1q` is supported; on models that only support 802.1Q and reject that command, omit that one line and keep `switchport mode trunk`.

4.2 SW1 — Sales access switch

```
enable
configure terminal
hostname SW1
no ip domain-lookup
vtp domain ABC-BRANCH
vtp password cisco123
vtp mode client

interface range fastEthernet0/1 - 2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
 no shutdown

interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,99
 no shutdown

! Shut down unused access ports; adjust range if ports are in use.
interface range fastEthernet0/3 - 24
 switchport mode access
 switchport access vlan 999
 shutdown
vlan 999
 name UNUSED
end
copy running-config startup-config
```

4.3 SW2 — Engineering access switch

```
enable
configure terminal
hostname SW2
no ip domain-lookup
vtp domain ABC-BRANCH
vtp password cisco123
vtp mode client

interface range fastEthernet0/1 - 2
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
 no shutdown

interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,99
 no shutdown

interface range fastEthernet0/3 - 24
 switchport mode access
 switchport access vlan 999
 shutdown
vlan 999
 name UNUSED
end
copy running-config startup-config
```

Important: VTP clients learn VLANs from the VTP server. The VLAN 999 unused-port VLAN should be created on SW-CORE as well if you want it propagated consistently; alternatively use a locally created unused VLAN on each switch. Add `vlan 999` on SW-CORE for a consistent VTP domain.

4.4 R1 — Branch router, inter-VLAN routing and DHCP

```
enable
configure terminal
hostname R1
no ip domain-lookup
ip domain-name abc.local
username admin privilege 15 secret Cisco123!
enable secret EnableCisco123!

interface gigabitEthernet0/0
 no shutdown

interface gigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 ip access-group BLOCK-SALES-TO-SERVER in

interface gigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip nat inside

interface gigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
 ip nat inside

interface gigabitEthernet0/1
 description WAN-to-R2
 ip address 200.1.1.1 255.255.255.252
 ip nat outside
 no shutdown

ip dhcp excluded-address 192.168.10.1 192.168.10.49
ip dhcp excluded-address 192.168.10.101 192.168.10.254
ip dhcp excluded-address 192.168.20.1 192.168.20.49
ip dhcp excluded-address 192.168.20.101 192.168.20.254

ip dhcp pool SALES-POOL
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 203.0.113.10

ip dhcp pool ENGG-POOL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 203.0.113.10

access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.99.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet0/1 overload

router ospf 1
 router-id 1.1.1.1
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.99.0 0.0.0.255 area 0
 network 200.1.1.0 0.0.0.3 area 0
 passive-interface gigabitEthernet0/0.10
```

```
 passive-interface gigabitEthernet0/0.20
 passive-interface gigabitEthernet0/0.99

ip access-list extended BLOCK-SALES-TO-SERVER
 deny ip 192.168.10.0 0.0.0.255 192.168.99.0 0.0.0.255
 permit ip any any

crypto key generate rsa general-key modulus 2048
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
end
copy running-config startup-config
```

The ACL is applied inbound on the Sales subinterface and denies Sales-to-server-farm traffic. The Engineering permit is not needed in this ACL because Engineering traffic enters R1 through Gi0/0.20, not Gi0/0.10. The final `permit ip any any` allows other routed traffic; add more specific rules if the business policy requires tighter isolation.

4.5 R2 — WAN / simulated external network router

```
enable
configure terminal
hostname R2
no ip domain-lookup
ip domain-name abc.local
username admin privilege 15 secret Cisco123!
enable secret EnableCisco123!

interface gigabitEthernet0/1
 description WAN-to-R1
 ip address 200.1.1.2 255.255.255.252
 no shutdown

interface gigabitEthernet0/0
 description Simulated-external-LAN
 ip address 203.0.113.1 255.255.255.0
 no shutdown

router ospf 1
 router-id 2.2.2.2
 network 200.1.1.0 0.0.0.3 area 0
 network 203.0.113.0 0.0.0.255 area 0
 passive-interface gigabitEthernet0/0

crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
end
copy running-config startup-config
```

Configure the external test server with IP 203.0.113.10, mask 255.255.255.0, gateway 203.0.113.1. Enable HTTP and, if DNS tests are required, DNS service with an appropriate record. Configure the internal server as 192.168.99.10/24, gateway 192.168.99.1.

4.6 ACL policy

Required behavior: Sales (192.168.10.0/24) cannot reach the server farm (192.168.99.0/24). Engineering (192.168.20.0/24) can reach the server farm. The ACL above enforces the Sales restriction. Verify with actual host traffic and ACL counters.

5. Verification Commands

Purpose	Command
VLANs	show vlan brief
Trunk state and allowed VLANs	show interfaces trunk
Interface status	show ip interface brief (routers); show interfaces status (switches)
VTP domain/mode	show vtp status
STP root	show spanning-tree vlan 10
Port security	show port-security interface fa0/1
DHCP leases	show ip dhcp binding
OSPF adjacency	show ip ospf neighbor
Routing table	show ip route
ACL hit counters	show access-lists BLOCK-SALES-TO-SERVER
NAT translations/statistics	show ip nat translations; show ip nat statistics
SSH status	show ip ssh; show users

5.1 Expected test matrix

Test	Expected result	Evidence to capture
Sales PC gets DHCP address	192.168.10.50–192.168.10.100	PC IP configuration; `show ip dhcp binding`
Engineering PC gets DHCP address	192.168.20.50–192.168.20.100	PC IP configuration; `show ip dhcp binding`
PC pings its gateway	Success	Ping output
Engineering PC → internal server 192.168.99.10	Success	Ping/HTTP output
Sales PC → internal server 192.168.99.10	Blocked	Ping/connection failure and ACL deny counter
R1 ↔ R2 OSPF	Neighbor FULL	`show ip ospf neighbor` on both routers
Internal PC → external test server 203.0.113.10	Success if server/gateway/services are configured	Ping/HTTP and NAT translation
Unauthorized MAC on secured access port	Port becomes err-disabled when violation is triggered	`show port-security interface fa0/1` and interface status

6. Troubleshooting Workflow

Ticket 001 — Sales cannot reach the external test server

1. Check PC addressing with `ipconfig` and confirm IP, mask, gateway and DNS.
2. Ping 192.168.10.1. If this fails, inspect PC access VLAN, trunk VLAN 10, R1 Gi0/0 and subinterface Gi0/0.10.
3. On R1, run `show ip interface brief`, `show interfaces trunk` on switches, and `show vlan brief`.
4. Check OSPF with `show ip ospf neighbor` and confirm R1 has a route to 203.0.113.0/24.
5. Check NAT with `show ip nat statistics` and `show ip nat translations` while generating traffic.
6. On R2, verify Gi0/0 is up and the external server uses 203.0.113.1 as its default gateway.
7. Retest HTTP/ping and document the actual root cause and output.

Ticket 002 — VLAN 20 missing on SW2

8. Run `show vtp status` on SW-CORE, SW1 and SW2; verify VTP version/domain/mode/password compatibility.
9. Run `show interfaces trunk` and confirm VLAN 20 is allowed and active on the SW-CORE–SW2 trunk.
10. Run `show vlan brief` on SW2 and confirm Fa0/1-2 are assigned to VLAN 20.
11. If the VLAN still does not appear, check the actual VTP mode/version and device model; alternatively configure VLANs locally and use transparent mode consistently.

7. Security and operational notes

- Replace example credentials before any real deployment. The credentials in this training document are not production-safe.
- Use SSH rather than Telnet for device management.
- Apply port security only to end-device access ports, not switch trunks or router uplinks.
- Shut down unused ports and place them in a dedicated unused VLAN.
- Use `copy running-config startup-config` after validating a change.
- Keep configuration backups and screenshots of real command output.
- VTP can propagate accidental VLAN changes; for production networks, use an explicitly planned VTP design or transparent mode where appropriate.

8. Known design limits

- This is a small lab, not a tested 200-user deployment. The addressing plan has capacity, but the physical scale is not represented by four PCs.
- The 203.0.113.0/24 network is a simulated external LAN. It does not provide real internet access.
- DNS works only if a DNS service and records are configured on the server at 203.0.113.10; otherwise use an available reachable DNS resolver for your lab.
- Router-on-a-stick is used for learning. A Layer 3 switch with SVIs may be more appropriate for some production designs.
- The exact interface names and support for some commands depend on the selected Packet Tracer/GNS3 device image.

10. Final validation checklist

- Physical topology matches the port map.
- VLANs exist and trunks carry VLANs 10, 20 and 99.
- All relevant interfaces are up/up.
- Sales and Engineering receive addresses from the correct DHCP pools.
- OSPF neighbor state is FULL and R1 learns 203.0.113.0/24.
- Engineering can access the internal server; Sales cannot.
- External test-server traffic creates a NAT/PAT translation.
- SSH login works with the local account.
- All PASS/FAIL statements in the final report are backed by captured output.